

Batch Normalization in Multi-Agent Traffic Signal Control RL

Number of project members: 1

Topic explored in class: Batch Normalization

Statistical Performance Evaluated: Variance of final avg loss across multiple random seeds

In this project, I implement a multi-agent reinforcement learning system for traffic signal control using the RESCO benchmark, specifically the Cologne single-intersection scenario ([cologne1](#)). Each traffic intersection is modeled as an agent, and a centralized training with decentralized execution framework is used, following the [Multi-Agent Actor-Critic for Mixed Cooperative-Competitive Environments](#) formulation. The baseline model consists of actor and critic networks implemented as feedforward neural networks, trained using an off-policy actor-critic method. The objective is defined by the RESCO reward formulation, and performance is evaluated using final average waiting time after a fixed number of training episodes. The primary focus of this project is training stability, defined as the variance of final average waiting time across random seeds.

The baseline is modified by introducing batch normalization (BN) into the network architecture. The comparison is between (1) the baseline model and (2) a variant with BN applied to the centralized critic. Prior work such as [Revisiting Batch Normalization for Training Low-Latency Deep Spiking Neural Networks From Scratch](#) shows that standard BN is insufficient when input distributions vary over time due to mismatch between normalization statistics and current inputs, and that introducing time-dependent normalization improves training stability. While their method explicitly models temporal variation, this project applies standard BN in a reinforcement learning setting, where non-stationarity arises from changing policies and replay buffer distributions. Based on this, the hypothesis is that BN applied to the centralized critic reduces variance across runs and improves final average waiting time by stabilizing value estimation under non-stationary training data.

The environment is provided by the [RESCO: An Efficient Reinforcement Learning Framework for Traffic Signal Control](#) benchmark, which includes SUMO-based simulations derived from real-world traffic networks. The cologne1 scenario is used to limit complexity while preserving realistic traffic dynamics. Observations and actions are defined per intersection, and joint information is used in the centralized critic.

Evaluation is conducted along two dimensions. Statistical performance is measured using final average waiting time. Training stability is measured as the standard deviation of final average waiting time across seeds.

Preliminary inspection of the RESCO framework indicates that agent-level observations and actions can be extracted and that the simulation loop supports replay-buffer-based training. This suggests that integrating the actor-critic model with BN is feasible within the project timeline.

The objective of this project is to quantify the effect of batch normalization on final performance and stability in a multi-agent actor-critic setting under non-stationary training conditions.

Experiment Plan

Hypothesis

Applying batch normalization to the centralized critic reduces variance across runs and improves final average waiting time under a fixed training budget by stabilizing value estimation under non-stationary data.

Metrics

1. Final average waiting time (task performance)
2. Variance of final average waiting time across seeds (stability)

Protocol

I will implement two versions of the same multi-agent actor-critic system:

1. A baseline without batch normalization
2. A variant with batch normalization applied to the centralized critic only.

Both models will be trained on the same cologne1 scenario for the same number of episodes and with the same optimizer, replay buffer settings, and network size. I will run each configuration with 3–5 random seeds. After training, I will record the final average waiting time for each run, compute the mean and standard deviation across seeds, and compare the total training time for the fixed budget.

Expected Results

The BN variant is expected to show lower variance across seeds and equal or lower final average waiting time.